



Ayşe Şimal Menekşe

📍 **Home** : Sundgauallee 42, 79110, Freiburg im Breisgau, Germany

✉ **Email**: aysesimalmenekse@gmail.com 📞 **Phone**: (+49) 15510880066

🌐 **LinkedIn**: [linkedin.com/in/simalmenekse](https://www.linkedin.com/in/simalmenekse) 🐙 **Github**: <https://github.com/simalmenekse>

Work permit: German **Date of birth**: 22/10/2003 **Nationality**: Turkish

EDUCATION AND TRAINING

[10/2021 – 06/2025]

Bachelor's Degree

Cankaya University

City: Ankara | **Country**: Türkiye | **Field(s) of study**: Computer Engineering | **Final grade**: GPA: 3.55

[08/2023 – 12/2023]

Bachelor's degree, Computer Science, Erasmus 2023

University of Vaasa

City: Vaasa | **Country**: Finland

[10/2025 – Current]

M.Sc. Computer Science

Albert-Ludwigs-Universität Freiburg, Germany

City: Freiburg | **Country**: Germany

Current GPA: **2.6**

Relevant Coursework: Machine Learning, Foundations of Deep Learning, Reinforcement Learning, Simulation in Computer Graphics

WORK EXPERIENCE

TÜBİTAK SAGE

City: Ankara | **Country**: Türkiye

[07/2023 – 08/2023]

Embedded Software Engineer Intern

- Developed embedded software components using C# and C++ for hardware systems
- Implemented UART communication protocols for serial data transmission

ASELSAN

City: Ankara | **Country**: Türkiye

[07/2024 – 08/2024]

Software Engineer Intern

- Built desktop application features using the QT Framework in C++
- Collaborated with engineering team on software testing and debugging

PUBLICATIONS

[2026]

[VRKitchen: AI-Guided Procedural Training and Risk Management in a Virtual Cooking Environment](#)

Developed an AI-driven VR training system integrating Unity and AWS Bedrock LLMs

Designed a model-driven approach for real-time procedural guidance and risk handling

Evaluated system usability with 14 participants, demonstrating effective interaction and robustness

Authors: Ayşe Şimal Menekşe, Zeynep Sila Mert, Tuna Yavuz, Işın Karakoç, Hüseyin Alperen Elbiz, Mert Kumbasar | **Journal Name**: 2026 5th International Informatics and Software Engineering Conference (IISEC), March 2026 | **Publisher**: IEEE

PROJECTS

[06/09/2023 – 12/2023]

Applied Machine Learning Project – Song Mood Prediction

Built a classification model to predict song mood from audio features using Python and Scikit-Learn during Erasmus semester at University of Vaasa.

[10/2024 – 06/2025]

VR-Kitchen: Risk Management Platform for Gastronomy Practitioners - Bachelor's Project

Led the development of a VR-based (Meta Quest 3) AI-powered cooking assistant that integrates Unity, Amazon Bedrock LLMs, and cloud APIs to deliver procedural training and real-time safety management.

Link: <https://github.com/CankayaUniversity/ceng-407-408-2024-2025-VRKitchen-Risk-Management-Platform-for-Gastronomy-Practitioners>

LANGUAGE SKILLS

Mother tongue(s): Turkish

Other language(s):

English

LISTENING C2 READING C2 WRITING C2

SPOKEN PRODUCTION C2 SPOKEN INTERACTION C2

German

LISTENING B1 READING B1 WRITING B1

SPOKEN PRODUCTION B1 SPOKEN INTERACTION B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Programming and Technical Skills

Keras | C / C# / C++ | Unity | AWS | LLMs | Computer Vision | Databases | SQL | Reinforcement Learning | Object-Oriented Programming (OOP) | Data Science | MongoDB | Python | Deep Learning | Machine Learning | RAG Systems | QT | Git | Data Analytics

NETWORKS AND MEMBERSHIPS

[2021 – 06/2025]

Member of IEEE Ankara

[2021 – 2025]

Member of Google Developer Student Clubs Ankara

HONOURS AND AWARDS

[08/2023]

IBM Applied AI Specialization Awarding institution: IBM / Coursera

[08/2023]

IBM Data Science Specialization Awarding institution: IBM / Coursera

[04/07/2023]

Google Advanced Data Analytics Specialization Awarding institution: Google / Coursera

CERTIFICATES

[18/11/2024 – Current]

IELTS Academic Score: 7.5